

Reproducible Regression Harness for Machine-Learning Security Tools

Contract-Validated Adapters Catching Cross-Repo Breakage Before It Ships

github.com/poojakira/mlsec-benchmark-suite (public, default branch main, primary language Python), MIT • Python 3.12 • HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60 • verified 2026-10-01

1.00

HF-SCANNER F1

3/0

TP / FP

83 across 10 modules

Test functions

82 passed, 1 skipped

Current CI

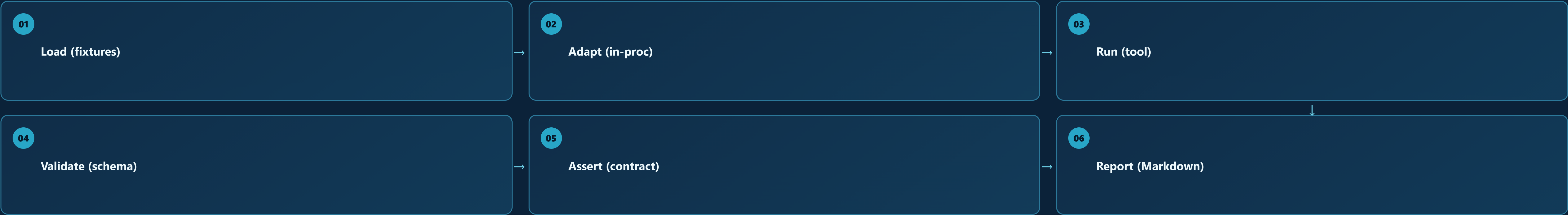
Contribution

A contract-validated regression harness that wraps ML security tools in typed in-process adapters, runs them against versioned fixtures, and fails pytest on schema drift - with a hard, labeled separation between REAL product adapters and a synthetic smoke-test scaffold.

Verified Evidence

- VERIFIED_AT_SNAPSHOT - Real HF-scanner adapter: precision=recall=F1=1.0 on 5 fixtures - README verified real run; tests/test_hf_scanner_adapter.py end-to-end (no mock); total_tp=3,total_fp=0,total_fn=0.
- VERIFIED_AT_SNAPSHOT - 83 test functions across 10 test modules; current Python 3.12 CI recorded 82 passed, 1 skipped with 90.07% statement coverage - repository README and test tree at HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60.
- VERIFIED_AT_SNAPSHOT - JSON-schema contracts + versioned fixtures; pytest fails on drift - README how-it-works; schemas/ + contracts/ present.
- UNSUPPORTED (disclaimed by repo) - run-smoke per-category numbers - README: seeded-PRNG synthetic scaffold for plumbing, NOT measurements. Shown as synthetic only.
- UNSUPPORTED (disclaimed) - Real-world accuracy of wrapped tools - Suite validates contracts/format, not accuracy.
- Statement coverage: 90.07%
- Real fixtures: 5
- Real adapters: gated CI

Architecture / Threat Flow



Flow labels come directly from the repository research brief. They describe the tested control path, not a universal security guarantee.

Methodology and Control Path

Load (fixtures)	Adapt (in-proc)
Run (tool)	Validate (schema)
Assert (contract)	Report (Markdown)

Limitations and Residual Risk

- Validates contracts/format, not real-world accuracy.
- run-smoke numbers are synthetic (seeded PRNG).
- Fixture coverage is finite; gaps possible.
- Adapters can go stale if a tool changes internals.
- Real adapter gating requires sibling install.

Reproducibility

```
git clone https://github.com/poojakira/mlsec-benchmark-suite.git
cd mlsec-benchmark-suite
git checkout 5c20f1da9602eb162efe8c63158da52b11d61f60
python -m pip install -e ".[dev]"
pytest tests/ -q
mlsec-benchmark run-hf-scanner --output r.json
```

Evidence: results/real_hf_scanner.json, METHODOLOGY.md, tests/